PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Global prevalence of dyslipidaemia in adult populations: a systematic review protocol
AUTHORS	Mohamed-Yassin, Mohamed-Syarif; Baharudin, Noorhida; Abdul-Razak, Suraya; Ramli, Anis Safura; Lai, Nai Ming

VERSION 1 – REVIEW

REVIEWER	Nabil Sulaiman University of Sharjah College of Medicine
REVIEW RETURNED	21-Mar-2021

GENERAL COMMENTS	This protocol for a systematic review " Global prevalence of dyslipidaemia in adult populations: a systematic review protocol" is much needed. The review objectives are clear and the methodology protocol is described in details. The planned statistics has been described in details but it me need review by senior statistician.

REVIEWER	Maria Naves Universidade Federal de Goiás
REVIEW RETURNED	28-May-2021

GENERAL COMMENTS	Suggestions for improvements in the protocol: - It is worth including as a limitation of the study, the possible lack of studies in specific regions of the globe, as South America or Australia. - line 113 – please include the meaning of ASCVD. - lines 118-119 – The statement “The World Health Organization’s estimates from 2008 showed that the prevalence of hypercholesterolemia was the highest in Europe…” could be replaced with a more current estimate? Is it about the adult population? - lines 144-146 – Why did the authors not include, in the search strategy, the following databases: PubMed (Medline was included but via Pubmed?), Web of Science, Scopus, and Elsevier-Embase? - line 146 – the authors did not limit the publication period of the articles for searching, but the data will be as useful as refer to a recent period of time, as the last 10 or 5 years. - lines 150-164 – please, clarify the search strategy detailing search terms and their combination in each database, using OR and AND. A Table is a recommended way of showing this information. - include the section “Inclusion and exclusion criteria” with the information of the lines 176-189, and remove the section “Type of study to be included”, avoiding repetition of information.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1 Dr. Nabil Sulaiman's Comments	Authors' Responses
This protocol for a systematic review "Global prevalence of dyslipidaemia in adult populations: a systematic review protocol" is much needed. The review objectives are clear and the methodology protocol is described in details. The planned statistics has been described in details but it me need review by senior statistician.	Thanks for the kind comments.
Reviewer 2 Dr. Maria Naves's Comments	**Authors' Responses**
Suggestions for improvements in the protocol: - It is worth including as a limitation of the study, the possible lack of studies in specific regions of the globe, as South America or Australia.	Thanks for your suggestions. -We have added "The main potential limitation of this systematic review could be the lack of studies on the prevalence of dyslipidaemia in specific regions around the world such as South America, Australia, and the Pacific Islands." under a new section entitled "Limitation of Study". (Page 9, lines 226-229)
- line 113 – please include the meaning of ASCVD.	Thanks for pointing this out. We have added the meaning of ASCVD: "atherosclerotic cardiovascular disease (ASCVD)." (Page 5, line 112)
- lines 118-119 – The statement "The World Health Organization's estimates from 2008 showed that the prevalence of hypercholesterolemia was the highest in Europe…" could be replaced with a more current estimate? Is it about the adult population?	Thanks for your comment. Unfortunately, we could not find a more current estimate. We hope the present systematic review that we are conducting will fill in this gap in data. Yes, the prevalence presented were in the adult population. We have added "in adults" to clarify this. (Page 5, line 118).
- lines 144-146 – Why did the authors not include, in the search strategy, the following databases: PubMed (Medline was included but via Pubmed?), Web of Science, Scopus, and Elsevier-Embase?	Thanks for the comment. We have added "PubMed/" before "MEDLINE" (Page 6, line 147). We did not include Web of Science, Scopus, and Elsevier-Embase in our search strategy as from our experience the databases that we have included (PubMed/MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL), which incorporates trials from Medline and EMBASE, as well as trial

	registry), cover most relevant articles to be screened, and additional databases such as Scopus and Web of Science yielded little additional relevant information amidst a large body of non-relevant articles. While we acknowledge that there is no best approach for screening that balances comprehensiveness against efficiency, we opted to focus on the databases with the highest relevant yields in view of our limitations in time and manpower. We acknowledge the possibility of potentially missing articles with our approach, and have included this as our limitations: "In view of our limitations in time and manpower, we have opted to focus on the databases with the highest relevant yields from our experience in our search strategy, which are PubMed/MEDLINE and CENTRAL. These databases cover most relevant articles to be screened and additional databases such as Web of Science and Scopus may yield little additional relevant information amidst a large body of non-relevant articles. Hence, there is a possibility of potentially missing articles with this approach." (Pages 9-10, lines 229-23).
- line 146 – the authors did not limit the publication period of the articles for searching, but the data will be as useful as refer to a recent period of time, as the last 10 or 5 years.	We will capture all relevant articles across all period for the sake of comprehensiveness. We will then look at the studies of each period to see if there is a clear difference.
- lines 150-164 – please, clarify the search strategy detailing search terms and their combination in each database, using OR and AND. A Table is a recommended way of showing this information.	Thanks for the suggestion. We have added "The main search strategies according to the different databases are shown in Table 1." (Page 6, line 150), and created "Table 1 Search strategies" to detail the search terms and their combinations, using the Boolean operators "OR" and "AND" where appropriate. (Page 6, line 152).

- include the section “Inclusion and exclusion criteria” with the information of the lines 176-189, and remove the section “Type of study to be included”, avoiding repetition of information.	Thanks for this suggestion. We have added a section entitled “Inclusion and exclusion criteria” and have moved the information from lines 176-189 to this new section: “We will include all cross-sectional and longitudinal observational studies reporting on the prevalence of elevated serum total cholesterol (TC), elevated low-density lipoprotein cholesterol (LDL-c), elevated triglycerides (TG), and low high-density lipoprotein cholesterol (HDL-c) in adults above 18 years old. We will exclude studies of children or individuals with familial hypercholesterolaemia in this review. We will also exclude editorials, commentaries, reviews, letters, case series with less than 50 patients, and studies without primary data or explicit description of methods. When a study is published in two or more reports, the most comprehensive one with the largest sample size will be included.” (Page 7, lines 158-165)

VERSION 2 – REVIEW

REVIEWER	Maria Naves Universidade Federal de Goiás
REVIEW RETURNED	09-Nov-2021

GENERAL COMMENTS	The authors reviewed the Manuscript ID bmjopen-2021-049662.R1 accordingly, but I would like to add one suggestion: In Table 1, it is not necessary to repeat the search strategy of item “A” in Item “B”, as the criteria for searching in the Pubmed database is the same of those of Central database.

VERSION 2 – AUTHOR RESPONSE

Reviewer 2 Dr. Maria Naves's Comments	Authors' Responses
The authors reviewed the Manuscript ID bmjopen-2021-049662.R1 accordingly, but I would like to add one suggestion: In Table 1, it is not necessary to repeat the search strategy of item "A" in Item "B", as the criteria for searching in the Pubmed database is the same of those of Central database.	Thanks for your suggestion. We have removed Item "B" and revised Item A's title to: "A. Search strategy PubMed/MEDLINE (adapted for CENTRAL)". Because of this change, we have also revised the title for Table 1 from "Search strategies" to "Search strategy" (Page 6, Line 152), and the sentence "The main search strategies according to the different databases are shown in Table 1." To "The main search strategy is shown in Table 1." (Page 6, Line 150).